VASIREDDY VENKATADRI INSTITUTE OF TECHNOLOGY

Approved by AICTE, Permanently Affiliated to JNTU, Kakinada

Accredited by NAAC with 'A' Grade - ISO 9001:2008 Certified

Nambur (V), Peda Kakani (M), Guntur Dt. - 522508 April, 2025.

A Project Report

on

AI Resume Analyser and Job Prediction

submitted for partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

Computer Science and Engineering

by

PULI ZENESWARI	**- 23BQ1A05I5**
PERISETLA TEJESWARARAO	**- 23BQ1A05H6**
POTHINA PAVAN KALYAN	**- 24BQ5A0519**
PETETI VAMSI KRISHNA	**-23BQ1A05H8**

Under the Guidance of

Dr. N. Lakshmi Prasanna

Professor, CSE

VASIREDDY VENKATADRI INSTITUTE OF TECHNOLOGY
(Autonomous)
Permanently Affiliated to JNTU, Kakinada, Approved by AICTE
Accredited by NAAC with 'A' Grade, ISO 9001:2015 Certified
Nambur, Pedakakani (M), Guntur (Dt) - 522508
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
B.Tech Programme Accredited by NBA

CERTIFICATE

This is to certify that the Project Report titled "**AI Resume Analyser and Job Prediction**" is being submitted by, **Ms. PULI ZENESWARI, Mr. PERISETLA TEJESWARARAO, Mr. POTHINA PAVAN KALYAN, Mr. PETETI VAMSI KRISHNA**, bearing Registered Numbers **23BQ1A05I5, 23BQ1A05H6, 24BQ5A0519, and 23BQ1A05H8** respectively of III B.Tech II semester Computer Science & Engineering is a record Bonafide work carried out by them. The results embodied in this report have not been submitted to any other University for the award of any degree.

Project Guide	**Head of Department**
(Dr. N. Lakshmi Prasanna, Professor)	(Dr. V. Ramachandran, Professor)

Submitted for Viva-voce Examination held on _____

Internal Examiner **External Examiner**

DECLARATION

We, **Ms. PULI ZENESWARI, Mr. PERISETLA TEJESWARARAO, Mr. POTHINA PAVAN KALYAN, Mr. PETETI VAMSI KRISHNA**, hereby declare that the Project Report entitled "Deep Learning – Traffic Violation Detection" done by us under the guidance of **Dr. N. Lakshmi Prasanna, Professor at Vasireddy Venkatadri Institute of Technology (A)** is submitted in partial fulfillment of the requirements for the award of Bachelor of Technology in Computer Science & Engineering. The results embodied in this report have not been submitted to any other university for the award of any degree**.**

DATE :

PLACE :

SIGNATURE OF THE CANDIDATE(s)

1.

2.

3.

4.

ACKNOWLEGMENT

We take this opportunity to express my deepest gratitude and appreciation to all those people who made this project work easier with words of encouragement, motivation, discipline, and faith by offering different places to look to expand our ideas and helped us towards the successful completion of this project work.

First and foremost, we express our deep gratitude to **Sri. Vasireddy Vidya Sagar**, Chairman, Vasireddy Venkatadri Institute of Technology (A) for providing necessary facilities throughout the B Tech program.

We express our sincere thanks to **Dr. Y. Mallikarjuna Reddy**, Principal, Vasireddy Venkatadri Institute of Technology (A) for his constant support and cooperation throughout the B. Tech program.

We express our sincere gratitude to **Dr. V. Ramachandran**, Professor & HOD, Computer Science & Engineering, Vasireddy Venkatadri Institute of Technology (A) for his constant encouragement, motivation and faith in offering different places to look to expand our ideas.

We would like to express our sincere gratefulness to our Guide **Dr. N. Lakshmi Prasanna**, Professor, CSE for his insightful advice, motivating suggestions, invaluable guidance, help and support in the successful completion of this project.

We would like to express our sincere heartfelt thanks to our Project Coordinator **Dr. Vidya Sagar Uppuluri**, Professor, CSE for his valuable advices, motivating suggestions, moral support, help and coordination among us in successful completion of this project.

We would like to take this opportunity to express our thanks to all the **Teaching and Non-Teaching** Staff in the Department of Computer Science & Engineering, VVIT for their invaluable help and support.

Names of Students

PULI ZENESWARI	23BQ1A05I5
PERISETLA TEJESWARARAO	23BQ1A05H6
POTHINA PAVAN KALYAN	24BQ5A0519
PETETI VAMSI KRISHNA	23BQ1A05H8

INSTITUTE VISSION

To impart quality education through exploration and experimentation and generate socially conscious engineers, embedding ethics and values, for the advancement in Science and Technology.

INSTITUTE MISSION

- To educate students with practical approach to dovetail them to industry needs.
- To govern the institution with a proactive and professional management with passionate teaching faculty.
- To provide holistic and integrated education and achieve over all development of students imparting scientific and technical, social and cognitive, managerial and organizational skills.
- To compete with the best and be the most preferred institution of the studious and the scholarly.
- To forge strong relationships and linkage with the industry.

DEPARTMENT VISSION

Providing quality education to enable the generation of socially conscious software engineers who can contribute to the advancement in the field of computer science and engineering.

DEPARTMENT MISSION

- To equip the graduates with the knowledge and skills required to enable them to be industry ready.
- To train socially responsible, disciplined engineers who work with good leadership skills and can contribute for nation building.
- To make our graduates proficient in cutting edge technologies through student centric teaching-learning process and empower them to contribute significantly to the software industry.
- To shape the department into a centre of academic and research excellence.

COURSE OUTCOMES

CO 1: Articulate problem statement. (K2)

CO 2: Apply technial knowledge. (K3)

CO 3: Acquire contemporaray tools & technologies. (K2)

CO 4: Communicate and present the entire SDLC. (K3)

CO 5: Perform the role of a team member or lead in SDLC. (K3)

Program Educational Objectives (PEOs)

The Programme Educational Objectives of the B.Tech in Computer Science Engineering programme are given below and are numbered from PEO1 to PEO4.

PEO1	To provide the graduates with solid foundation in computer science and engineering along with fundamentals of Mathematics and Sciences with a view to impart in them high quality technical skills like modelling, analyzing, designing, programming and implementation with global competence and helps the graduates for life-long learning.
PEO2	To prepare and motivate graduates with recent technological developments related to core subjects like programming, databases, design of compilers and Network Security aspects and future technologies so as to contribute effectively for Research & Development by participating in professional activities like publishing and seeking copy rights.
PEO3	To train graduates to choose a decent career option either in high degree of employability / Entrepreneur or, in higher education by empowering students with ethical administrative acumen, ability to handle critical situations and training to excel in competitive examinations.
PEO4	To train the graduates to have basic interpersonal skills and sense of social responsibility that paves them a way to become good team members and leaders.

Program Outcomes (POs)

The B.Tech CSE program has documented measurable outcomes that are based on the needs of the program's stakeholders. The program outcomes, which are derived from ABET criteria, were first drafted in the academic year 2009–10 and later revised in 2010–11. The program outcomes that the department presently adopts are as follows:

1	**Engineering knowledge**	Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
2	**Problem analysis**	Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural science and engineering sciences.
3	**Design/development of solutions**	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations.
4	**Conduct investigations of complex problems**	Use research based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5	**Modern tool usage**	Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6	**The Engineer and Society**	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7	**Environment and Sustainability**	Understand the impact of the professional engineering solutions in the societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8	**Ethics**	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9	**Individual and Team Work**	Function effectively as an individual and as a member or leader in diverse teams, and in multidisciplinary settings.
10	**Communication**	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11	**Project Management and Finance**	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12	**Lifelong Learning**	Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

PSO-1	**Professional Skills:**	The ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, multimedia, web design, big data analytics and networking for efficient design of computer-based systems of varying complexity.
PSO-2	**Successful Career and Entrepreneurship:**	The ability to employ modern computer languages, environments, and platforms in creating innovative career paths to be an entrepreneur and a zest for higher studies/employability in the field of Computer Science & Engineering.

CO – PO ARTICULATION MATRIX

Course Outcomes:

CO	Description
CO1	Articulate problem statement. (K2)
CO2	Apply technical knowledge. (K2)
CO3	Acquire contemporary tools & technologies. (K2)
CO4	Communicate and present the entire SDLC. (C3)
CO5	Perform the role of a team member or lead in SDLC. (K3)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	1			2	2	3		2			1	
CO2	2	2	3	2	2		2	1	2	2	2	1	3	1
CO3		1	1	1	3							1	1	
CO4	1								3	3			1	1
CO5		1						2	3	3	2		1	2

INDEX

CH No	TITLE	PAGE NO
CHAPTER 1	**INTRODUCTION**	1–3
	1.1 Introduction	
CHAPTER 2	**AIM AND SCOPE**	4-8
	2.1 Existing System	
	2.2 Proposed System	
CHAPTER 3	**CONCEPT AND METHODS**	9-16
	3.1 Problem Description	
	3.2 Proposed Solution	
	3.2.1 Dataset	
	3.2.2 Data Preparation	
	3.2.3 Data Preprocessing	
	3.2.4 Data Augmentation	
	3.2.5 System Models/Working Mechanism	
	3.2.6 Performance Metrics	
CHAPTER 4	**IMPLEMENTATION**	17-31
	4.1 Tools Used	
	4.2 Code	
	4.3 Results and Screenshots	
CHAPTER 5	**CONCLUSION AND FUTURE SCOPE**	32-34
	REFERENCES	
	APPENDIX	
	Conference Publication Certificate(s)	
	Published Article in the Journal / Proceedings	

CHAPTER – 1

INTRODUCTION

In today's competitive world, technology plays a vital role in shaping career opportunities. Many industries expect graduates to possess not only academic knowledge but also practical technical skills. Companies look for candidates who are proficient in programming languages, development tools, databases, and emerging technologies such as Artificial Intelligence and Data Science. However, many students complete their education without having a clear understanding of the skills required for specific job roles. This creates a gap between academic learning and industry expectations, making it difficult for students to prepare effectively for placements and career growth.

Students often face confusion while choosing the right career path. For example, a student interested in the software field may not know whether to focus on web development, mobile application development, artificial intelligence, cloud computing, or data analytics. Without proper guidance, students may spend time learning irrelevant technologies that do not contribute to their career goals. As a result, they may lack confidence during interviews and job applications.

Most students depend on multiple platforms such as job portals, online learning websites, blogs, social media, and video tutorials to understand industry requirements. Although these platforms provide valuable information, the guidance is often scattered and not personalized. Students need to search different sources to identify the skills required for a particular job role. This process is time-consuming and may lead to confusion because different sources provide different recommendations. Moreover, many platforms do not analyze the student's individual skill level before suggesting career options.

To solve these challenges, the AI Resume Analyzer and Job Prediction System is proposed. This system is designed to help students understand their current skill level and identify suitable job roles based on their resume. The platform uses Artificial Intelligence and Machine Learning techniques to analyze resume content and extract important information such as technical skills, certifications, education details, and project experience.

The system performs skill extraction using Natural Language Processing (NLP) techniques. It scans the uploaded resume and identifies the technologies and competencies mentioned by the student. After extracting the skills, the system compares

them with the requirements of different job roles available in the database. Based on this comparison, the platform predicts the most suitable job roles and provides a confidence score indicating how well the student matches each role.

Another important feature of the system is skill gap analysis. This process helps students understand which skills they already possess and which skills they need to improve. For example, if a student aims to become a Data Analyst but lacks knowledge of tools such as Python, SQL, or data visualization techniques, the system highlights these missing skills and provides suggestions for improvement. This helps students follow a structured learning path instead of randomly selecting topics to study.

The platform also evaluates the authenticity of skills by providing simple assessments or quizzes related to the mentioned technologies. This ensures that students not only claim skills in their resume but also demonstrate practical knowledge. Based on assessment results, the system generates personalized recommendations to enhance employability.

Additionally, the system provides a career guidance module that suggests learning resources, certification programs, and project ideas relevant to the predicted job roles. This helps students improve their professional profile and prepare effectively for interviews. The platform can be extended to support multiple domains such as software development, data science, cybersecurity, and cloud computing.

The proposed system is developed using modern web technologies to ensure efficiency and usability. The frontend is designed using React JS to provide an interactive and user-friendly interface. The backend services are implemented using Python with the Flask framework to handle resume processing, skill analysis, and job prediction logic. Machine Learning models are used to generate accurate job recommendations, and databases such as MongoDB or Firebase are used to store user profiles, skill data, and prediction results. Cloud deployment tools like GitHub and Render are used to host and manage the application.

Overall, the AI Resume Analyzer and Job Prediction System provides a centralized and intelligent platform to support students in career planning. By combining resume analysis, job prediction, skill gap identification, and personalized learning recommendations, the system helps students align their skills with industry requirements. This platform can significantly reduce confusion in career decision-making and improve students' confidence, employability, and professional growth.

CHAPTER – 2

AIM AND SCOPE

The project titled **"AI Resume Analyzer and Job Prediction System"** aims to develop an intelligent web-based platform that helps students analyze their resumes and identify suitable job roles based on their skills. In today's competitive job market, industries expect graduates to possess practical technical knowledge along with academic qualifications. However, many students are uncertain about the exact skills required for different job roles such as Software Developer, Web Developer, Data Analyst, Artificial Intelligence Engineer, or Cloud Engineer. Due to this lack of awareness and proper guidance, students often face challenges during campus placements and job applications. This project aims to address this issue by providing a centralized platform where students can upload their resumes, evaluate their skills, and receive career guidance based on industry requirements.

The main objective of this system is to reduce the gap between the skills acquired through academic learning and the skills expected by industries. The platform analyzes the content of student resumes using Artificial Intelligence and Natural Language Processing techniques. Based on the extracted skills, the system predicts suitable job roles and provides a confidence score that indicates how well the student matches each role. This helps students clearly understand their current career readiness level.

Another important goal of the system is to provide personalized skill improvement suggestions. The platform performs skill gap analysis by comparing the student's existing skills with the skills required for a selected job role. If certain required skills are missing, the system highlights them and recommends learning resources, certifications, or project ideas. This structured guidance enables students to focus on relevant technologies instead of learning random topics from multiple sources.

In addition, the system also supports skill authenticity verification through simple assessments or quizzes. This feature ensures that students can validate their practical knowledge and improve their confidence while attending interviews. The platform also generates detailed

reports that help students track their progress and plan their career development more effectively.

The proposed system is implemented as a web-based application so that students and administrators can access it easily from any location using a web browser. The user interface is designed to be simple, interactive, and user-friendly, enabling students to upload resumes, view job predictions, and follow learning recommendations without difficulty. By integrating Artificial Intelligence technologies with modern web development tools, the platform provides an efficient solution for career planning and employability enhancement.

Overall, the **AI Resume Analyzer and Job Prediction System** provides a smart and centralized solution for helping students make informed career decisions. By combining resume analysis, job role prediction, skill gap identification, and personalized learning guidance, the system supports students in improving their technical competencies and aligning their skills with industry expectations.

2.1 EXISTING SYSTEMS

At present, students rely on various platforms such as job portals, resume review tools, online learning websites, career blogs, and social media to understand the requirements of different job roles. Although these platforms provide useful information, they do not offer an integrated solution that can automatically analyze resumes and provide personalized job predictions or career guidance.

1. Manual Resume Evaluation and Career Guidance

In many educational institutions, resume evaluation and career guidance are provided through faculty members, placement trainers, or career counseling sessions. Students receive suggestions about resume preparation, interview skills, and required technical knowledge. However, this traditional approach has several limitations:

- Resume feedback may not be detailed or personalized for every student
- Limited availability of career mentors for individual guidance
- Students may still be unsure about suitable job roles based on their skills

2. Online Resume Builder and Learning Platforms

Many students use online resume builder tools and learning platforms to improve their profiles and acquire new skills. These platforms provide templates, tutorials, and certification courses. However, they do not help students clearly understand which job roles match their current skill set.

Some limitations include:

• Resume tools focus only on formatting rather than skill analysis
• Learning platforms provide generic content without personalized recommendations
• Students may learn technologies that are not directly relevant to their career goals

3. Job Portals and Career Websites

Job portals provide job listings along with descriptions of required qualifications and skills. Students can use this information to understand industry expectations. However, these platforms mainly focus on recruitment and job applications rather than career readiness analysis.

4. Skill Assessment Platforms

Some websites offer aptitude tests, coding challenges, or technical assessments. These platforms help students evaluate their performance in specific subjects. However, they do not combine assessment results with resume analysis to provide complete career guidance.

5. Limitations of Existing Systems

Even though the current systems provide useful resources for career preparation, they still have several drawbacks:
• Career guidance information is scattered across multiple platforms
• Students must spend more time searching for relevant resources
• Absence of personalized learning and career recommendations

Due to these limitations, students often struggle to identify suitable job roles and plan their career preparation effectively.

2.2 PROPOSED SYSTEM

To overcome the limitations of existing systems, this project proposes the AI Resume Analyzer and Job Prediction System, an intelligent web-based platform that helps students analyze their resumes and identify suitable job roles based on their skills. The system provides automated resume analysis, job prediction, skill gap identification, and personalized career guidance through a single centralized platform.

1. Resume Skill Extraction

The primary feature of the proposed system is automatic resume analysis. Students can upload their resumes in digital format, and the system uses Natural Language Processing techniques to extract important details such as technical skills, educational qualifications, certifications, and project experience. This process helps in understanding the student's current competency level without manual evaluation.

2. Job Role Prediction

After extracting skills from the resume, the system compares them with industry job role requirements stored in the database. Based on this comparison, the platform predicts suitable job roles such as Software Developer, Web Developer, Data Analyst, or Machine Learning Engineer. The system also provides a confidence score that indicates how closely the student's profile matches each job role.

3. Skill Gap Identification and Recommendations

The system performs skill gap analysis by identifying missing or weak skills required for the predicted job roles. It then provides recommendations such as learning resources, certification suggestions, and project ideas to help students improve their employability. This structured guidance enables students to focus on relevant technologies and follow a clear career preparation path..

4. Skill Authenticity Evaluation

To ensure that students possess practical knowledge of the skills mentioned in their resumes, the platform includes basic assessment features such as quizzes or technical questions. Based

on assessment performance, the system validates skill authenticity and provides further improvement suggestions.

5. System Implementation

The proposed system is implemented using modern web technologies to ensure efficiency and scalability. The frontend is developed using React JS to provide an interactive and user-friendly interface. The backend services are implemented using Python with the Flask framework to handle resume processing, machine learning predictions, and data management. Databases such as MongoDB or Firebase are used to store user profiles, resume data, job role information, and prediction results.

6. Benefits of the Proposed System

The AI Resume Analyzer and Job Prediction System offers several advantages:

- Provides intelligent resume analysis and job role prediction
- Performs automatic skill gap identification
- Offers personalized career guidance and learning recommendations
- Helps students understand their career readiness level
- Supports structured career planning and placement preparation

By integrating Artificial Intelligence, resume analysis, and career recommendation features into a single platform, the proposed system helps students align their skills with industry requirements and make better career decisions. This system can significantly improve student confidence, employability, and professional development.

CHAPTER – 3

CONCEPTS AND METHODS

3.1 PROBLEM DESCRIPTION

In today's competitive job market, students are expected to possess not only academic knowledge but also practical technical skills that match industry requirements. However, many students face difficulties in identifying the exact skills needed for different job roles. This situation creates a gap between the skills students currently have and the competencies expected by employers.

Most students depend on multiple scattered resources such as job portals, online learning platforms, resume-building websites, and career guidance blogs to understand job requirements. Although these platforms provide useful information, they usually offer general suggestions rather than personalized guidance based on a student's actual resume content and skill level. As a result, students may spend valuable time learning technologies that are not directly aligned with their desired career path.

Another major challenge is the lack of an intelligent system that can automatically analyze a student's resume and provide targeted career recommendations. Students often do not have a clear understanding of their strengths and weaknesses in relation to specific job roles such as Software Developer, Data Analyst, Web Developer, or Machine Learning Engineer. Without proper analysis, it becomes difficult for them to plan their career preparation effectively.

In addition, traditional methods of resume evaluation and career counseling require manual effort and may not provide consistent feedback for every student. This can lead to confusion and reduced confidence during placement preparation. Students need a smart solution that can quickly evaluate their profiles and guide them toward suitable career opportunities.

Therefore, the main problem addressed in this project is the absence of a centralized intelligent platform that can analyze resumes, predict suitable job roles, identify skill gaps, and provide structured career guidance in a single system.

The **AI Resume Analyzer and Job Prediction System** aims to solve this problem by offering an automated solution that helps students understand their current skill level, match their profiles with industry job roles, and receive personalized recommendations for skill improvement. This system supports students in making informed career decisions and preparing effectively for their professional future.

.

3.2 PROPOSED SOLUTION

To overcome the challenges identified in the problem description, the AI Resume Analyzer and Job Prediction System proposes an integrated intelligent web-based solution that helps students analyze their resumes, evaluate their career readiness, and improve their employability.

The proposed system consists of several modules that work together to provide a comprehensive career guidance and job prediction environment.

First, the system allows students to create an account and upload their resumes in digital format. The platform extracts important details such as personal information, educational qualifications, technical skills, certifications, and project experience using Natural Language Processing techniques. This information forms the basis for evaluating the student's suitability for different job roles.

Next, the system maintains a database containing various job roles such as Software Developer, Data Analyst, Web Developer, Machine Learning Engineer, and Cloud Engineer along with their required skill sets. Based on the skills extracted from the resume, the system automatically compares the student's profile with job role requirements.

The platform then performs Job Role Prediction and Skill Gap Analysis, where the student's existing skills are matched with the required skills of suitable job roles. The system identifies missing skills and calculates a confidence score that indicates how well the student's profile aligns with each job role.

After identifying the missing skills, the system generates Personalized Career Recommendations, which include suggestions for learning new technologies, completing

relevant certifications, and working on practical projects. This structured guidance helps students focus on improving the most important skills required for their desired career path.

Another important component of the system is the Skill Authenticity Evaluation module, which provides basic assessments or quizzes to verify the practical knowledge of students in the skills mentioned in their resumes. Based on performance, the system provides additional improvement suggestions.

By integrating these features into a single platform, the proposed system provides students with:

• Automatic resume analysis and job prediction
• Clear identification of missing or weak skills
• Personalized learning and career improvement guidance
• Better understanding of career readiness level
• Support for structured placement preparation

Thus, the AI Resume Analyzer and Job Prediction System acts as an intelligent career assistant that helps students make informed decisions, enhance their technical competencies, and achieve their professional goals.

.

3.2.1 DATASET

The dataset used in the **AI Resume Analyzer and Job Prediction System** consists of information related to student resumes, job roles, and the technical skills required for different positions in the industry. The dataset plays a crucial role in performing resume analysis, predicting suitable job roles, and identifying skill gaps for career improvement.

The primary dataset includes the following components:

Student Resume Dataset

This dataset contains information extracted from the resumes uploaded by students who use the system. It includes:

• Candidate ID
• Name
• Email ID
• Educational Qualifications
• Technical Skills
• Certifications
• Project Experience

This information helps the system understand the candidate's profile and evaluate their suitability for different job roles.

Job Role Dataset

This dataset contains details about various job roles available in the IT and technology industry. Each job role includes:

• Job Role ID
• Job Role Title
• Required Skills
• Job Role Description

Examples of job roles considered in the system include:

• Software Developer
• Web Developer
• Data Analyst
• Machine Learning Engineer
• Cloud Engineer

Skill Dataset

This dataset contains a list of important technical skills required for different job roles. Some examples include:

• Programming Languages – Java, Python, C, C++
• Web Technologies – HTML, CSS, JavaScript
• Database Technologies – MySQL, MongoDB
• Frameworks and Tools – React JS, Flask, Git, TensorFlow

The system compares the skills extracted from the student's resume with the required skill set

of various job roles. Based on this comparison, the platform predicts suitable job roles, calculates confidence scores, and generates skill gap reports to help students improve their career readiness.

3.2.2 DATA PREPARATION

Data Preparation

Data preparation is an important stage in the AI Resume Analyzer and Job Prediction System to ensure that the information used for resume analysis and job prediction is accurate, structured, and suitable for processing.

The following steps are involved in data preparation:

Data Collection

Information related to job roles, required technical skills, and industry expectations is collected from reliable sources such as job portals, company career pages, industry reports, and placement guidelines. In addition, student resume data is collected when users upload their resumes to the system.

Data Structuring

The collected data is organized into structured formats such as database tables and datasets. Each dataset contains specific attributes such as candidate ID, extracted skills, job role ID, required skills, and prediction results. Structuring the data helps the system process information efficiently and perform accurate comparisons between student profiles and job requirements.

Data Cleaning

In this step, unnecessary or incorrect data is identified and removed to improve system performance. For example:

• Duplicate skill entries extracted from resumes are eliminated
• Incorrect or outdated job role information is corrected
• Missing or incomplete data values are handled using appropriate techniques

This process ensures that only relevant and reliable data is used for analysis.

Data Storage

After preparation, the processed data is stored in a database such as MongoDB or Firebase. This allows the system to quickly retrieve resume information, job role datasets, and skill details during job prediction and skill gap analysis.

Proper data preparation helps the platform function efficiently and improves the accuracy of resume analysis, job role prediction, and career recommendation results.

3.2.3 DATA PREPROCESSING

Data processing in the AI Resume Analyzer and Job Prediction System involves analyzing the information extracted from the student's resume and comparing it with the skill requirements of different job roles. This process helps the system predict suitable job opportunities and provide career improvement suggestions.

The system performs the following operations:

Skill Extraction and Matching

The system first extracts technical skills, certifications, and experience details from the uploaded resume using Natural Language Processing techniques. These extracted skills are then compared with the required skills stored in the job role dataset.

Skill Gap Identification

Any skill that is required for a particular job role but not found in the student's resume is considered a missing skill. The system identifies these gaps and highlights them as areas where the student needs improvement.

Confidence Score Calculation

The system calculates a confidence score or match percentage that indicates how closely the student's profile matches the requirements of a specific job role.

Example:

Total Required Skills for a Job Role = 10
Skills Found in Student Resume = 6

Confidence Score = (6 / 10) × 100 = **60%**

This score helps students understand their current level of career readiness.

Job Prediction and Report Generation

Based on the skill matching results, the system predicts suitable job roles and generates a detailed report that includes:

• Predicted Job Roles
• Confidence Score for each role
• Missing or weak skills
• Suggestions for skill improvement

The processed data enables students to clearly understand their strengths and weaknesses and take appropriate steps to enhance their employability.

3.2.4 DATA AUGMENTATION

Data augmentation in the AI Resume Analyzer and Job Prediction System refers to enhancing and expanding the existing datasets to improve the accuracy and effectiveness of resume analysis and job prediction results.

This process involves adding additional relevant information and updating system data based on changing industry requirements.

It includes the following activities:

• Adding new job roles and updating job descriptions as industry demands evolve
• Updating the list of required technical skills based on emerging technologies and frameworks
• Including multiple career domains such as software development, data science, cloud computing, and artificial intelligence
• Enhancing the dataset by integrating learning resource links, certification suggestions, and project recommendations

By continuously improving and updating the dataset, the system remains relevant to current industry trends and provides more accurate career guidance to students. This helps users receive better job predictions and more useful skill improvement recommendations while preparing for modern career opportunities.

3.2.5 SYSTEM MODELS/WORKING MECHANISMS

The working mechanism of the **AI Resume Analyzer and Job Prediction System** follows a systematic and structured process that helps students analyze their resumes and prepare for suitable job roles.

1. User Registration and Login

Students create an account on the platform by providing basic details such as name, email, and password. After successful registration, users can log in securely to access system features.

2. Resume Upload and Profile Creation

After logging in, students upload their resumes in digital format. The system automatically extracts important information such as educational qualifications, technical skills, certifications, and project experience to create the user profile.

3. Resume Analysis and Skill Extraction

The system uses Natural Language Processing techniques to analyze the uploaded resume and identify relevant technical skills and competencies mentioned by the student.

4. Job Role Prediction

Based on the extracted skills, the system compares the student's profile with the job role dataset and predicts suitable job roles such as Software Developer, Data Analyst, Web Developer, or Machine Learning Engineer.

5. Skill Gap Identification and Confidence Score Generation

The platform identifies missing skills required for the predicted job roles and calculates a confidence score or match percentage that indicates how well the student's profile matches industry requirements.

6. Career Recommendation and Report Generation

The system generates a detailed report that includes predicted job roles, confidence scores, missing skills, and suggestions for improvement such as recommended technologies, certifications, or project ideas.

7. Skill Assessment and Improvement Guidance

Students may attempt basic assessments to validate their technical knowledge. Based on performance, the system provides further guidance to improve practical skills and career readiness.

This workflow enables students to systematically analyze their resumes, understand their strengths and weaknesses, and take appropriate steps to enhance their employability.

3.2.6 PERFORMANCE METRICS:

Performance metrics are used to evaluate how effectively the **AI Resume Analyzer and Job Prediction System** analyzes student resumes and provides accurate job predictions and career guidance.

The following metrics are considered:

Skill Matching Accuracy

This metric measures how accurately the system identifies and matches the technical skills extracted from a student's resume with the required skills of different job roles. Higher accuracy ensures better job prediction results and more reliable career recommendations.

Prediction Response Time

This metric evaluates how quickly the system processes the uploaded resume and generates job role predictions, confidence scores, and skill gap reports. Faster response time improves user experience and system efficiency.

User Satisfaction

This metric assesses how helpful students find the platform in understanding their career readiness, identifying missing skills, and receiving personalized improvement suggestions. Feedback from users helps in improving system features and performance.

System Reliability

This metric ensures that the platform operates consistently without errors, crashes, or data loss. A reliable system allows users to upload resumes, view predictions, and access reports smoothly at any time.

These performance metrics help in analyzing the overall effectiveness, usability, and accuracy of the AI Resume Analyzer and Job Prediction System in supporting student career development.

CHAPTER – 4

IMPLEMENTATION

4.1 TOOLS USED

The development of the AI Resume Analyzer and Job Prediction System requires various software tools and technologies to design, implement, and manage the system effectively. These technologies help in building the user interface, processing resume data, performing job prediction, and maintaining system performance.

The major tools used in the implementation of the project are described below.

1. Programming Language

Python

Python is used to implement the core functionalities of the system such as resume analysis, skill extraction, machine learning model integration, and job prediction logic.

Python helps in:

• Processing resume data
• Implementing Natural Language Processing algorithms
• Performing skill matching and prediction
• Generating career recommendation reports

2. Frontend Technologies

React JS

React JS is used to design and develop the user interface of the system. It helps in creating interactive and responsive web pages that allow students to upload resumes, view predictions, and access reports easily.

HTML (Hypertext Markup Language)

HTML is used to define the structure of web pages, including forms, navigation menus, tables, and content sections.

CSS (Cascading Style Sheets)

CSS is used to enhance the appearance of the user interface by designing layouts, colors, fonts, and responsive page structures.

JavaScript

JavaScript is used to add dynamic features such as form validation, real-time interaction, and data handling on the client side.

3. Backend Technology

Flask Framework

Flask is used as the backend framework to handle system operations such as user authentication, resume processing, machine learning predictions, and communication with the database.

The backend is responsible for:

• Managing user login and registration
• Processing resume uploads
• Performing job prediction and skill gap analysis
• Handling system requests and responses

4. Database

MongoDB / Firebase

The database is used to store and manage system data such as:

• Student profile information
• Resume details
• Job role datasets

• Skill requirements
• Prediction results and reports

The database ensures secure storage and efficient retrieval of information.

5. Development Environment

Visual Studio Code

This development environment is used for writing, debugging, and testing the project code efficiently. It provides features such as:

• Code highlighting
• Debugging support
• Extension integration
• Project management tools

6. Version Control System

Git and GitHub

Git is used to track changes in the project code and manage different versions. GitHub is used to store the project repository and support collaboration among team members.

7. Web Browser

Web browsers such as Google Chrome, Mozilla Firefox, or Microsoft Edge are used to run and test the web-based application to ensure proper functionality and user experience.

4.2 CODE STRUCTURE

The AI Resume Analyzer and Job Prediction System is developed using a modular architecture that separates different functionalities into independent components. This modular structure improves system maintainability, scalability, and ease of future enhancements.

The main modules in the system include:

1. User Authentication Module

This module manages the security and access control features of the platform. It handles:

• User registration
• Secure login and logout
• Session management
• Role-based access control

This ensures that only authorized users can upload resumes and access prediction results.

2. Resume Management Module

This module allows students to upload and manage their resumes. It performs the following functions:

• Uploading resumes in supported formats
• Storing resume data securely
• Extracting profile information such as education, skills, and experience

The information collected in this module is used for further analysis and job prediction.

3. Job Role Management Module

This module maintains the database of different job roles and their required skill sets. Administrators can:

• Add new job roles
• Update required skills based on industry trends
• Manage job role descriptions

This module ensures that the system uses updated job requirement data for prediction.

4. Resume Analysis and Job Prediction Module

This is the core component of the system. It performs the following operations:

• Extracts technical skills using Natural Language Processing
• Compares extracted skills with job role requirements
• Predicts suitable job roles
• Calculates confidence scores or match percentages
• Generates detailed prediction reports

5. Skill Gap Recommendation Module

This module identifies missing or weak skills required for predicted job roles and provides improvement suggestions. It includes:

• List of missing skills
• Suggested certifications or courses
• Recommended project ideas

This helps students enhance their employability in a structured manner.

6. Skill Assessment Module

This module provides basic quizzes or technical questions to verify the authenticity of the skills mentioned in the resume. Based on assessment results, the system offers further guidance for skill development.

7. Report Generation Module

This module generates user-friendly reports that include:

• Predicted job roles
• Confidence scores
• Skill gap analysis results
• Career improvement suggestions

Students can view or download these reports for future reference.

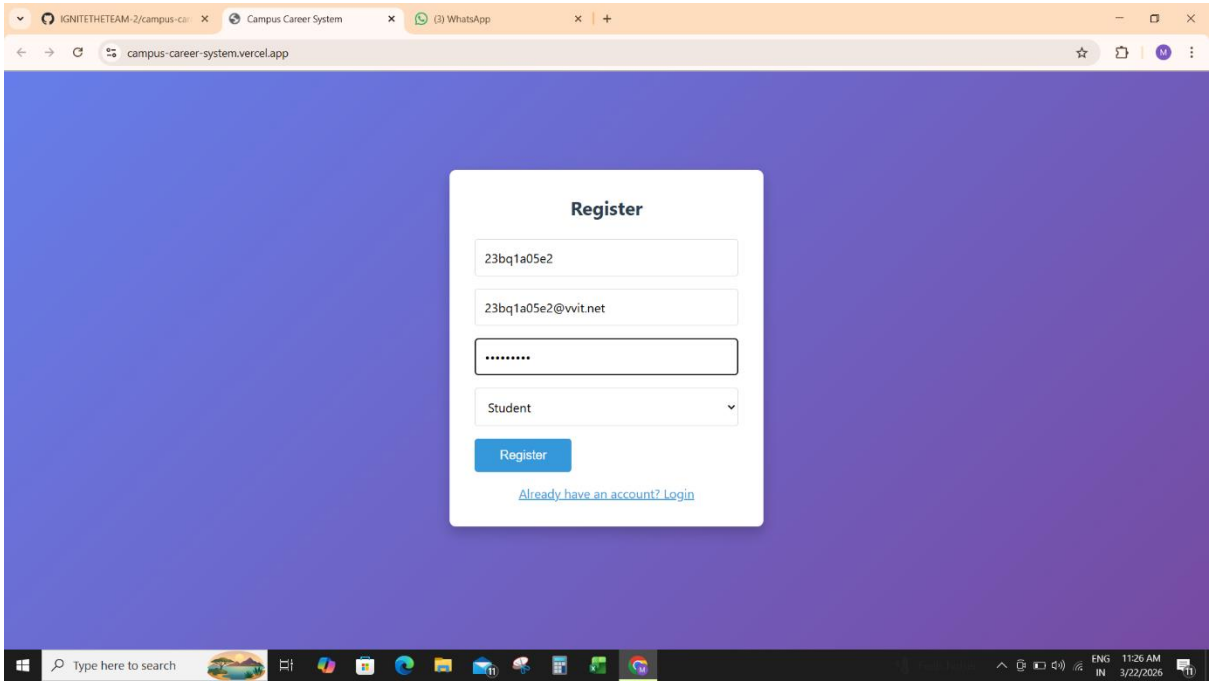
.

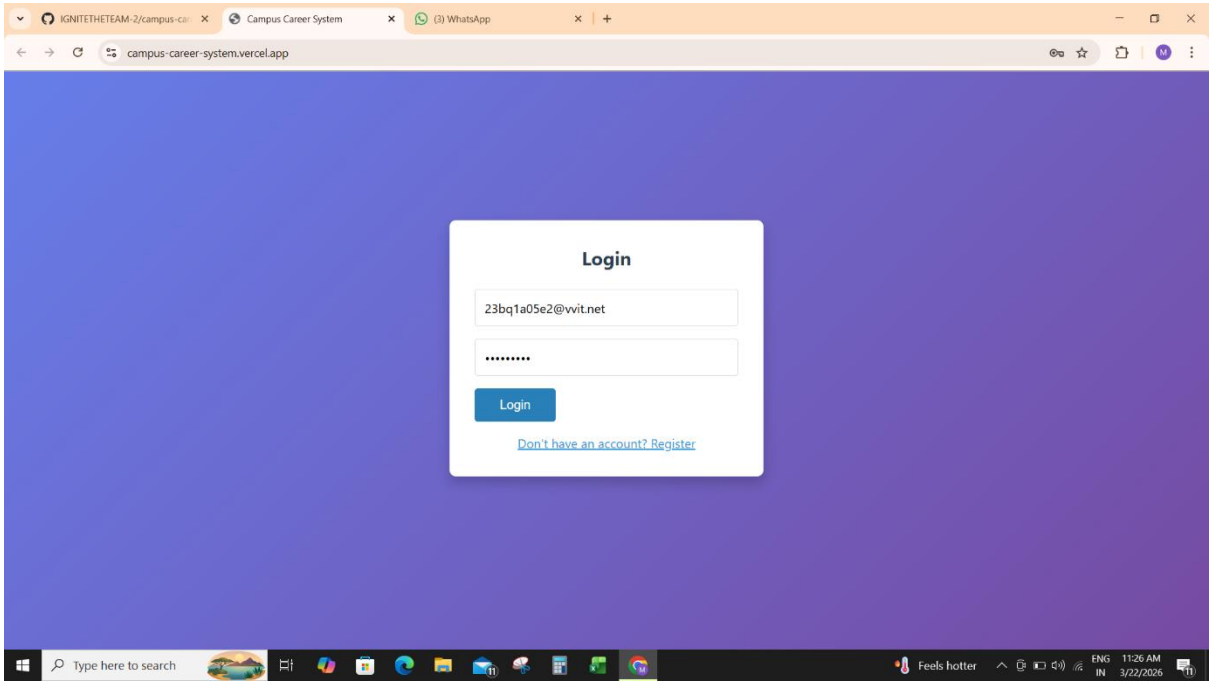
4.3 RESULTS AND OUTPUT SCREENSHOTS

After implementing the system, the **Smart Career Readiness Platform** successfully performs the following operations:

1. User Registration and Login

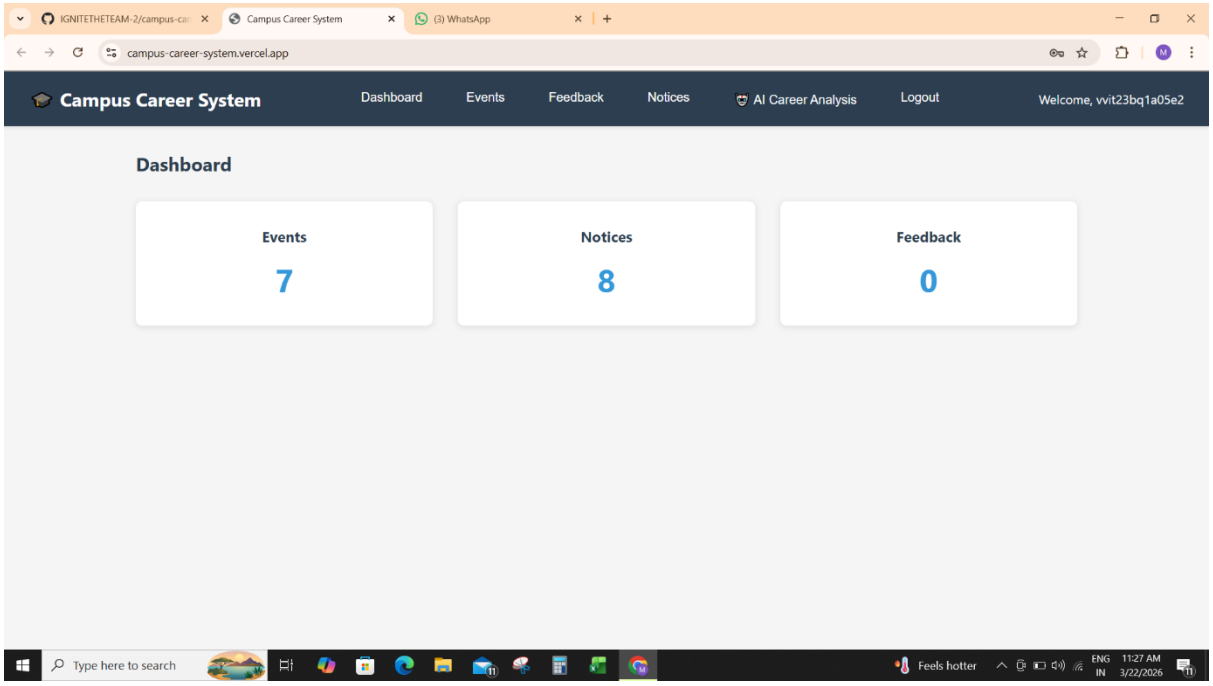
Students can create an account and log in securely to the platform.

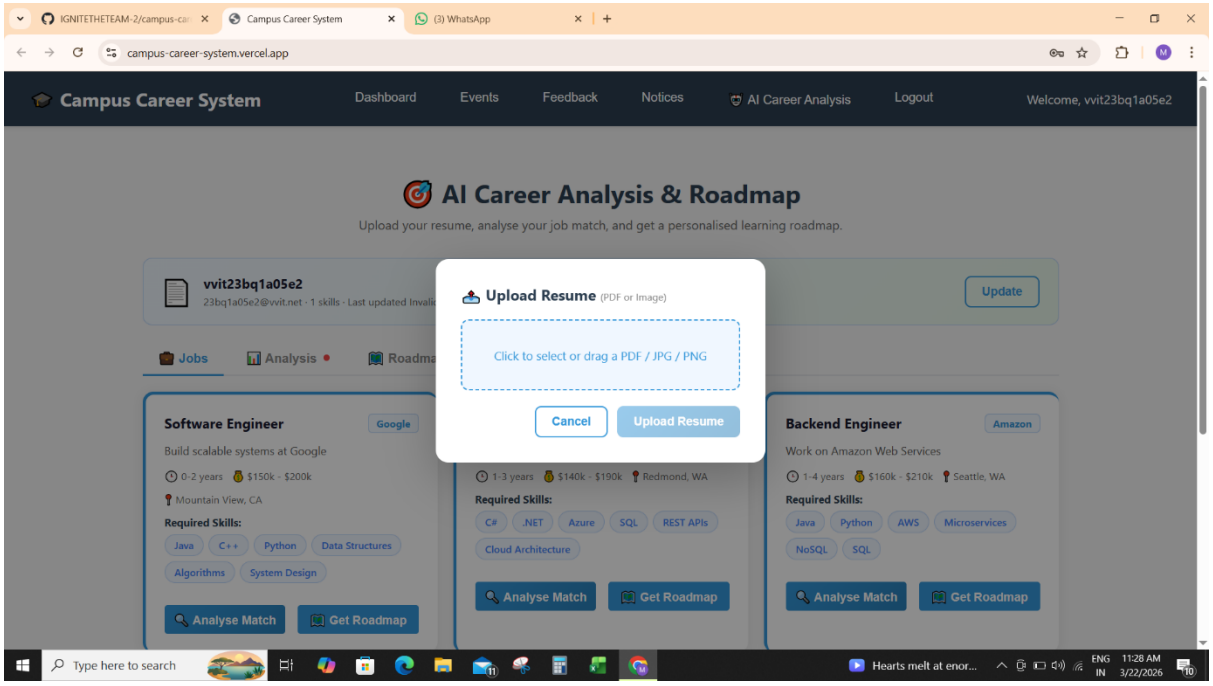




2. Profile Creation

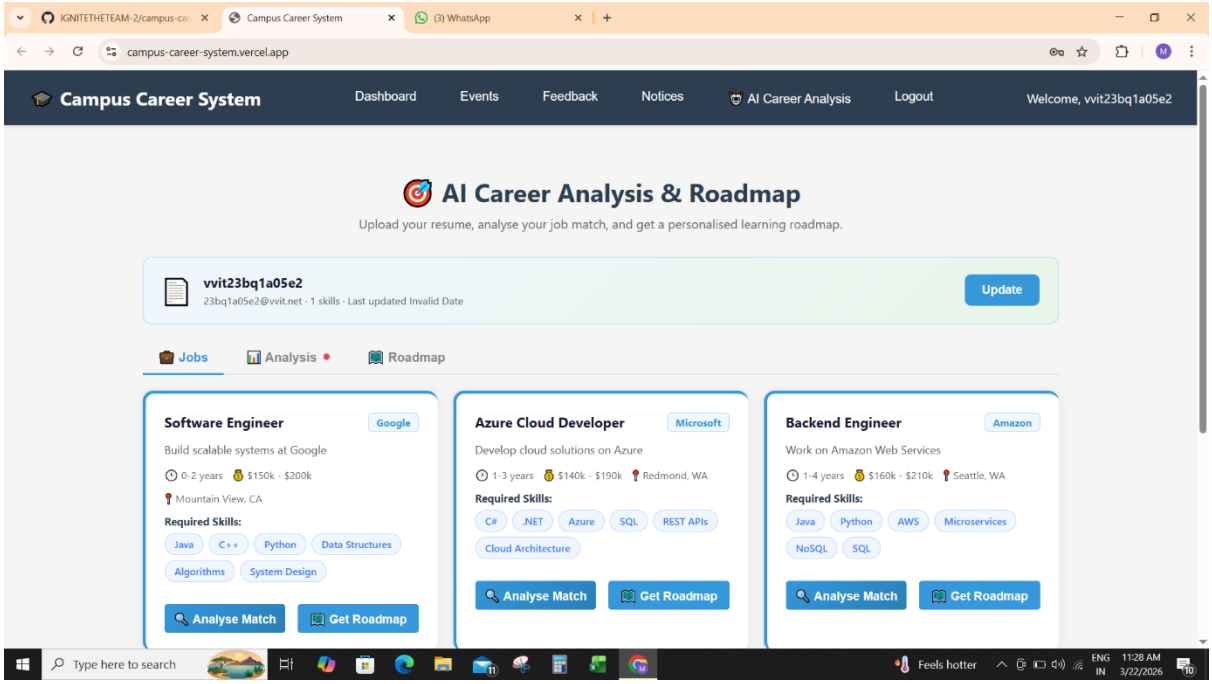
Students can enter their academic details and current skill sets.





3. Job Role Selection

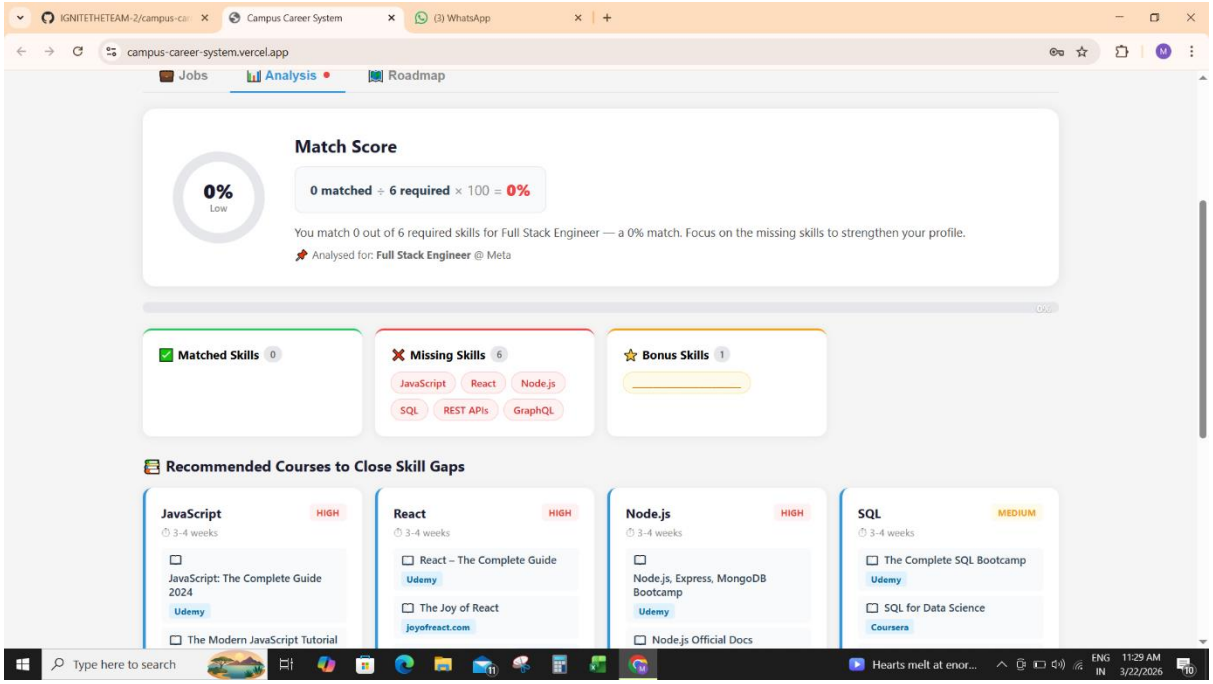
Students can select their desired career role from the available list of job roles.

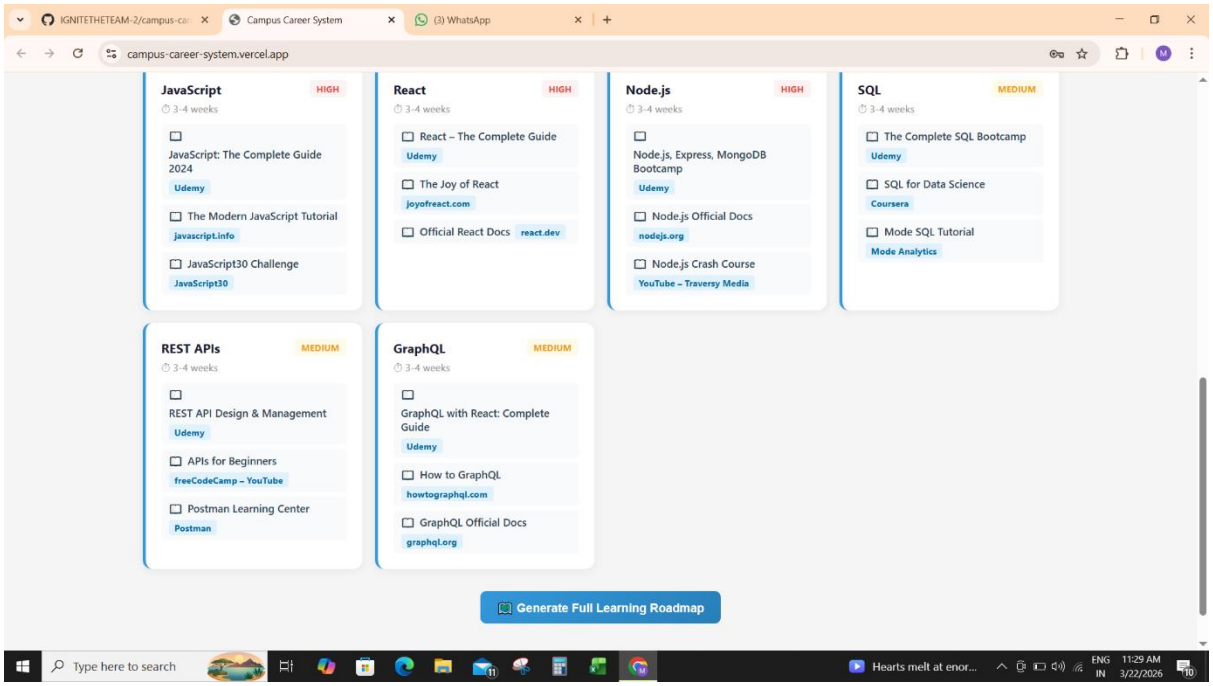


4. Skill Gap Report Generation

The system compares student skills with job role requirements and generates a report that includes:

- Missing skills
- Match percentage
- Recommendations for improvement





5. Learning Roadmap Generation

The system provides a step-by-step learning plan that helps students acquire the necessary skills for their chosen career path.

Start GitHub portfolio

3

Portfolio & Advanced Practice

4-6 weeks

Master skills through real-world projects and build a strong portfolio

Portfolio Projects | System Design | Code Quality

Courses & Activities:

- System Design for Software Engineers — Educative
- Clean Code Fundamentals — Pluralsight
- Build an end-to-end capstone project
- Write technical blog posts or case studies

Resources:

System Design Primer (GitHub) | Educative.io System Design Path (Educative) | Clean Code Book (Robert C. Martin) | Tech With Tim Project Tutorials (YouTube)

Milestones:

- Deploy capstone project
- Publish 2 case studies

4

Interview Preparation

2-3 weeks

Prepare for technical and behavioural interviews

Data Structures & Algorithms | System Design | Behavioural Interviews

Courses & Activities:

2

Intermediate Skill Development

6-8 weeks

Deepen knowledge in: SQL, REST APIs, GraphQL

SQL | REST APIs | GraphQL

Courses & Activities:

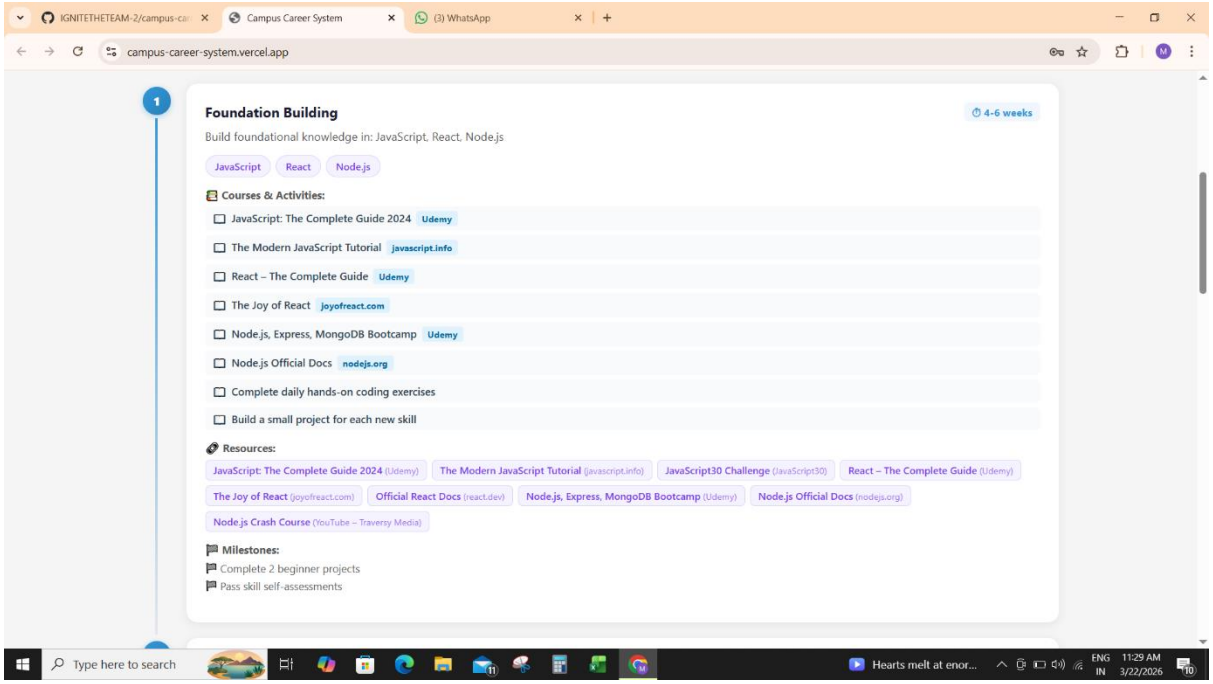
- The Complete SQL Bootcamp — Udemy
- SQL for Data Science — Coursera
- REST API Design & Management — Udemy
- APIs for Beginners — freeCodeCamp – YouTube
- GraphQL with React: Complete Guide — Udemy
- How to GraphQL — howtographql.com
- Build 2-3 portfolio projects combining new skills
- Contribute to open-source repositories on GitHub

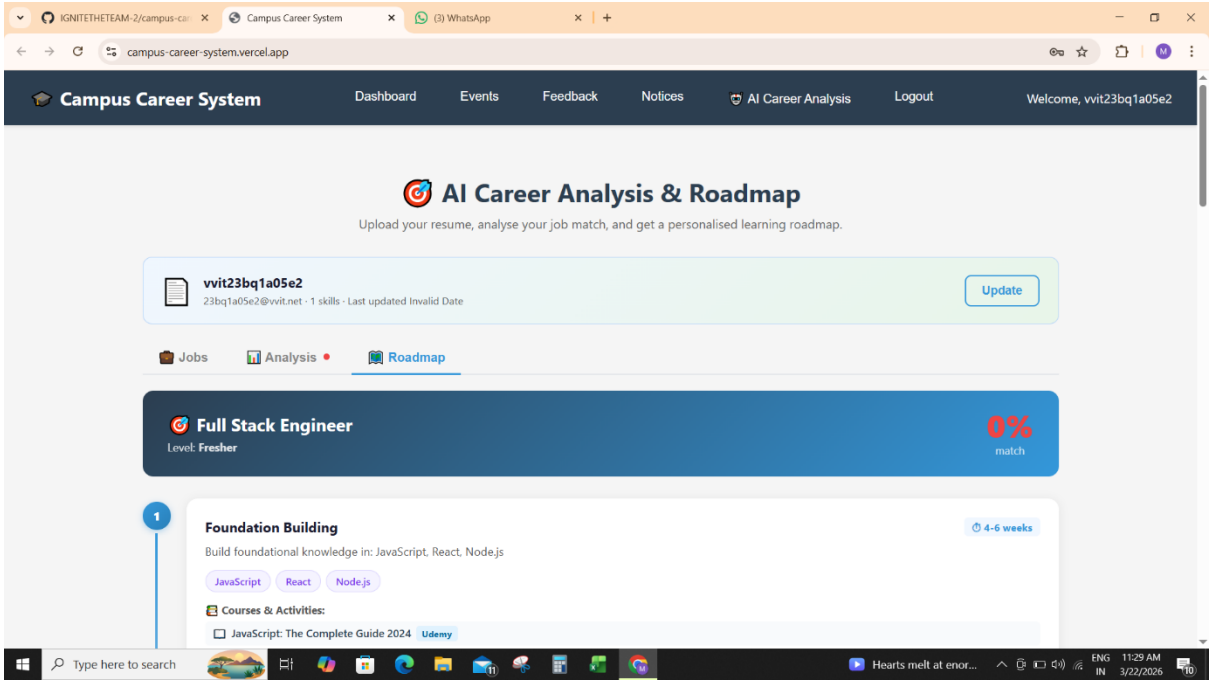
Resources:

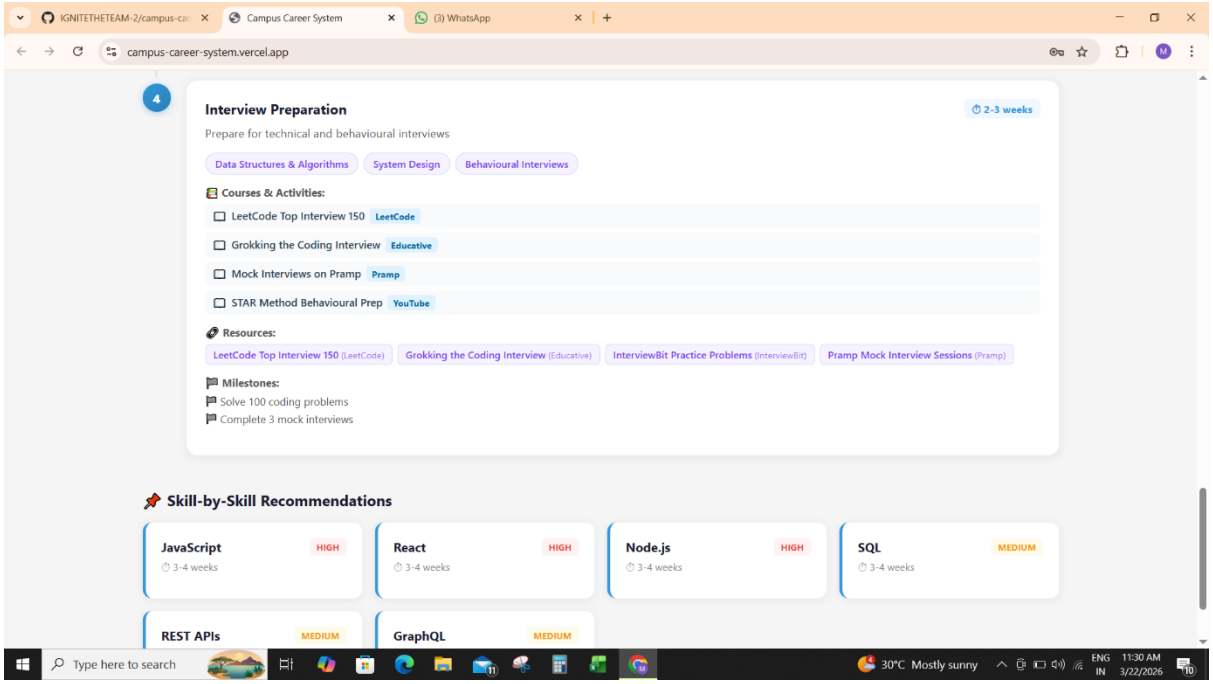
The Complete SQL Bootcamp (Udemy) | SQL for Data Science (Coursera) | Mode SQL Tutorial (Mode Analytics) | REST API Design & Management (Udemy) | APIs for Beginners (freeCodeCamp – YouTube) | Postman Learning Center (Postman) | GraphQL with React: Complete Guide (Udemy) | How to GraphQL (howtographql.com) | GraphQL Official Docs (graphql.org)

Milestones:

- Complete 3 intermediate projects
- Start GitHub portfolio


IGNITETHETEAM-2/campus-ca
Campus Career System
(3) WhatsApp
campus-career-system.vercel.app
1
Foundation Building
4-6 weeks
Build foundational knowledge in: JavaScript, React, Node.js
JavaScript
React
Node.js
Courses & Activities:
JavaScript: The Complete Guide 2024 Udemy
The Modern JavaScript Tutorial javascript.info
React – The Complete Guide Udemy
The Joy of React joyofreact.com
Node.js, Express, MongoDB Bootcamp Udemy
Node.js Official Docs nodejs.org
Complete daily hands-on coding exercises
Build a small project for each new skill
Resources:
JavaScript: The Complete Guide 2024 (Udemy)
The Modern JavaScript Tutorial (javascript.info)
JavaScript30 Challenge (JavaScript30)
React – The Complete Guide (Udemy)
The Joy of React (joyofreact.com)
Official React Docs (react.dev)
Node.js, Express, MongoDB Bootcamp (Udemy)
Node.js Official Docs (nodejs.org)
Node.js Crash Course (YouTube – Traversy Media)
Milestones:
Complete 2 beginner projects
Pass skill self-assessments
Type here to search
Hearts melt at enor...
ENG IN
11:29 AM 3/22/2026


IGNITETHETEAM-2/campus-ca
Campus Career System
(3) WhatsApp
campus-career-system.vercel.app
Campus Career System
Dashboard
Events
Feedback
Notices
AI Career Analysis
Logout
Welcome, vvit23bq1a05e2
AI Career Analysis & Roadmap
Upload your resume, analyse your job match, and get a personalised learning roadmap.
vvit23bq1a05e2
23bq1a05e2@vvit.net · 1 skills · Last updated Invalid Date
Update
Jobs
Analysis
Roadmap
Full Stack Engineer
Level: Fresher
0%
match
1
Foundation Building
4-6 weeks
Build foundational knowledge in: JavaScript, React, Node.js
JavaScript
React
Node.js
Courses & Activities:
JavaScript: The Complete Guide 2024 Udemy
Type here to search
Hearts melt at enor...
ENG IN
11:29 AM 3/22/2026

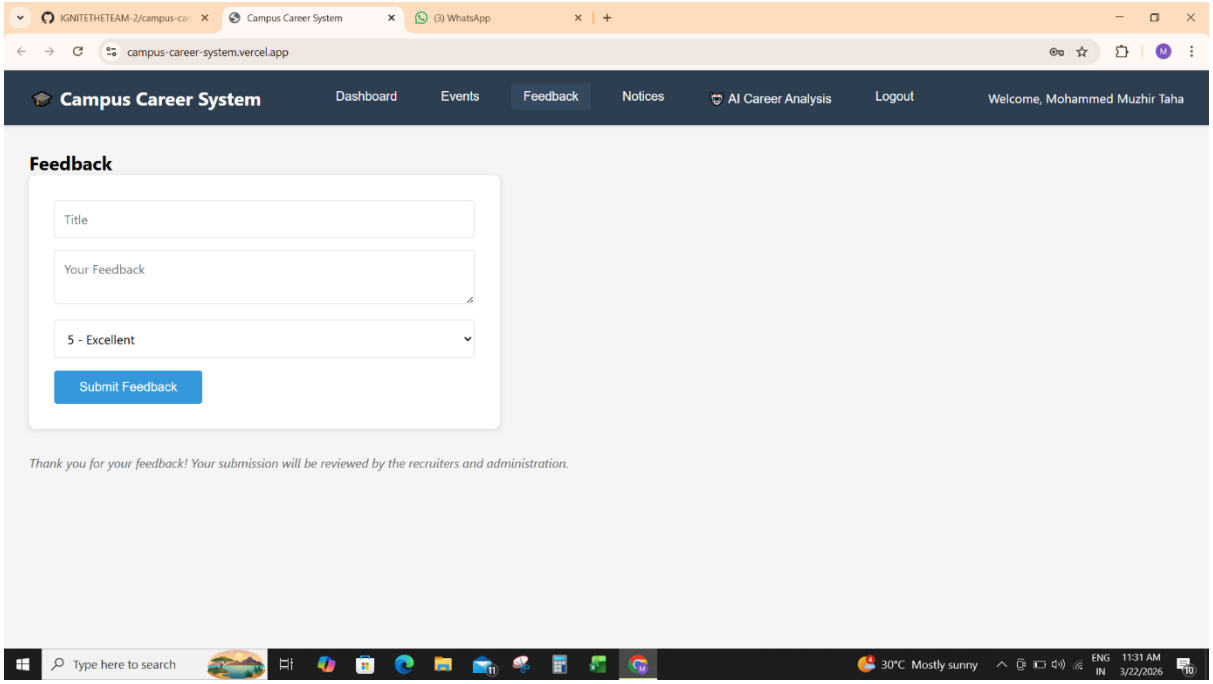


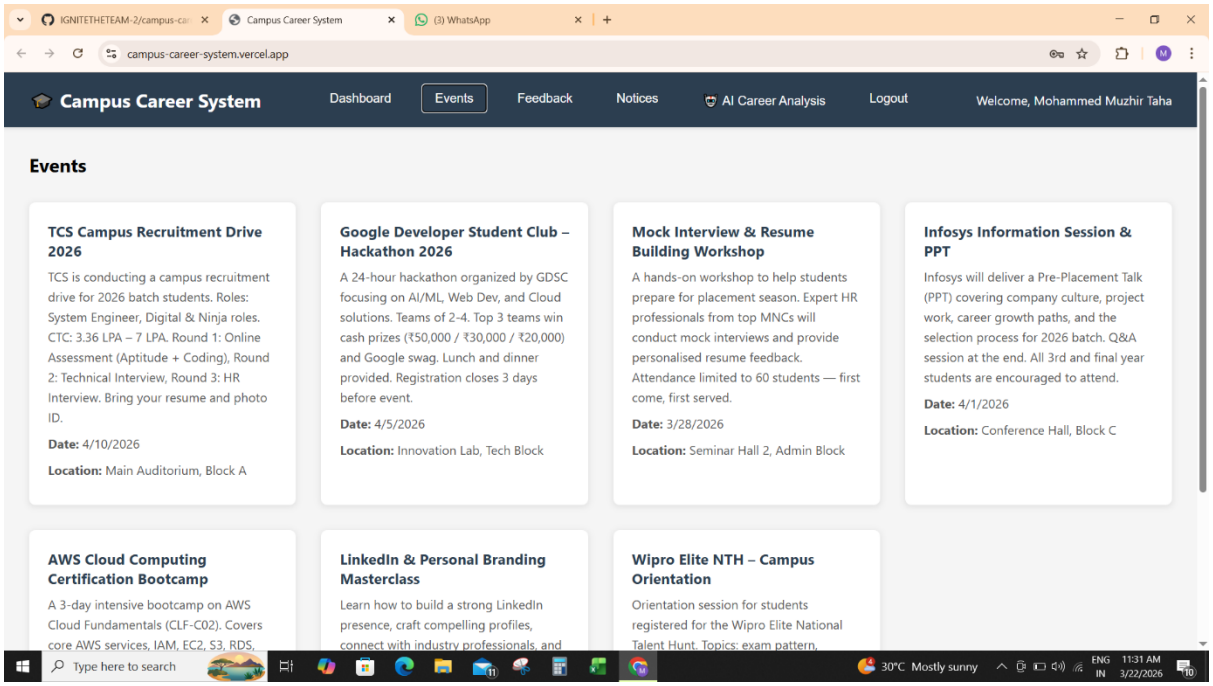
6. Digital Notice Board

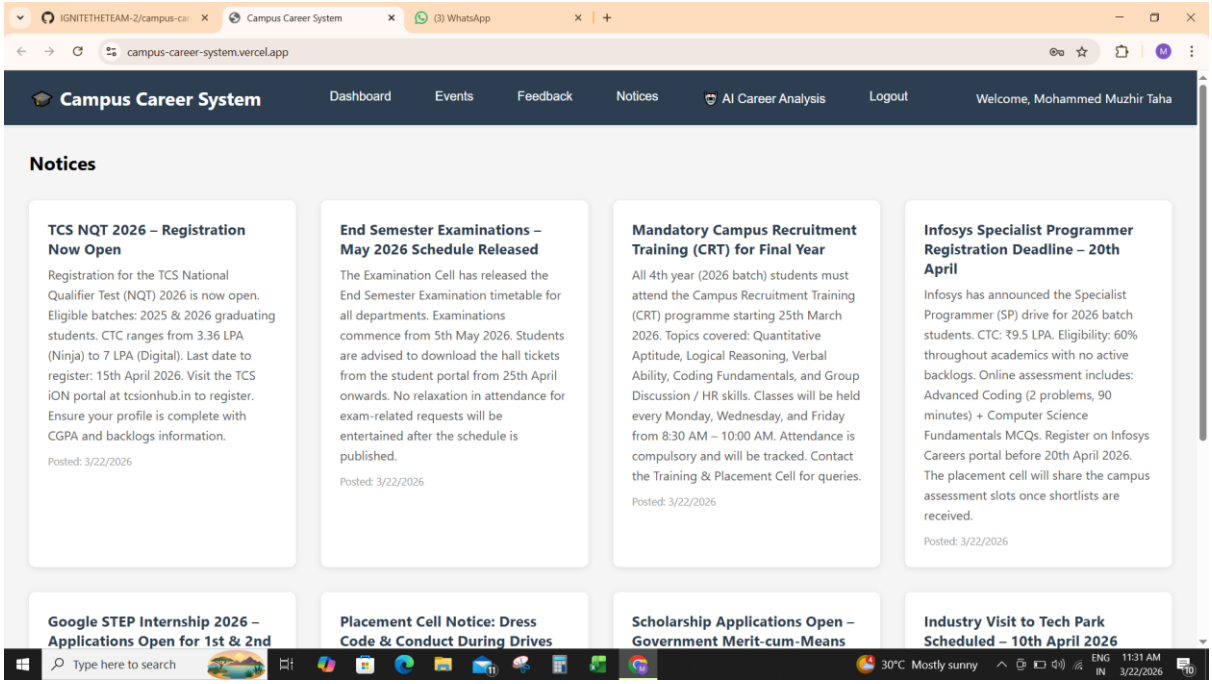
Students can view important updates posted by administrators related to placements, training sessions, and career opportunities.

The results demonstrate that the system effectively helps students understand their current skill level and provides guidance to improve their career readiness.

Screenshots of the system interface, including the login page, dashboard, skill gap report, and notice board, are included to illustrate the working of the system.


IGNITETHETEAM-2/campus-ca
Campus Career System
(3) WhatsApp
campus-career-system.vercel.app
Campus Career System
Dashboard
Events
Feedback
Notices
AI Career Analysis
Logout
Welcome, Mohammed Muzhir Taha
Feedback
Title
Your Feedback
5 - Excellent
Submit Feedback
Thank you for your feedback! Your submission will be reviewed by the recruiters and administration.
Type here to search
30°C Mostly sunny
ENG IN
11:31 AM 3/22/2026


IGNITETHETEAM-2/campus-ca
Campus Career System
(3) WhatsApp
campus-career-system.vercel.app
Campus Career System
Dashboard
Events
Feedback
Notices
AI Career Analysis
Logout
Welcome, Mohammed Muzhir Taha
Events
TCS Campus Recruitment Drive 2026
TCS is conducting a campus recruitment drive for 2026 batch students. Roles: System Engineer, Digital & Ninja roles. CTC: 3.36 LPA – 7 LPA. Round 1: Online Assessment (Aptitude + Coding), Round 2: Technical Interview, Round 3: HR Interview. Bring your resume and photo ID.
Date: 4/10/2026
Location: Main Auditorium, Block A
Google Developer Student Club – Hackathon 2026
A 24-hour hackathon organized by GDSC focusing on AI/ML, Web Dev, and Cloud solutions. Teams of 2-4. Top 3 teams win cash prizes (₹50,000 / ₹30,000 / ₹20,000) and Google swag. Lunch and dinner provided. Registration closes 3 days before event.
Date: 4/5/2026
Location: Innovation Lab, Tech Block
Mock Interview & Resume Building Workshop
A hands-on workshop to help students prepare for placement season. Expert HR professionals from top MNCs will conduct mock interviews and provide personalised resume feedback. Attendance limited to 60 students — first come, first served.
Date: 3/28/2026
Location: Seminar Hall 2, Admin Block
Infosys Information Session & PPT
Infosys will deliver a Pre-Placement Talk (PPT) covering company culture, project work, career growth paths, and the selection process for 2026 batch. Q&A session at the end. All 3rd and final year students are encouraged to attend.
Date: 4/1/2026
Location: Conference Hall, Block C
AWS Cloud Computing Certification Bootcamp
A 3-day intensive bootcamp on AWS Cloud Fundamentals (CLF-C02). Covers core AWS services, IAM, EC2, S3, RDS,
LinkedIn & Personal Branding Masterclass
Learn how to build a strong LinkedIn presence, craft compelling profiles, connect with industry professionals, and
Wipro Elite NTH – Campus Orientation
Orientation session for students registered for the Wipro Elite National Talent Hunt. Topics: exam pattern,
Type here to search
30°C Mostly sunny
ENG IN
11:31 AM 3/22/2026


IGNITETHETEAM-2/campus-ca
Campus Career System
(3) WhatsApp
campus-career-system.vercel.app
Campus Career System
Dashboard
Events
Feedback
Notices
AI Career Analysis
Logout
Welcome, Mohammed Muzhir Taha
Notices
TCS NQT 2026 – Registration Now Open
Registration for the TCS National Qualifier Test (NQT) 2026 is now open. Eligible batches: 2025 & 2026 graduating students. CTC ranges from 3.36 LPA (Ninja) to 7 LPA (Digital). Last date to register: 15th April 2026. Visit the TCS iON portal at tcsionhub.in to register. Ensure your profile is complete with CGPA and backlogs information.
Posted: 3/22/2026
End Semester Examinations – May 2026 Schedule Released
The Examination Cell has released the End Semester Examination timetable for all departments. Examinations commence from 5th May 2026. Students are advised to download the hall tickets from the student portal from 25th April onwards. No relaxation in attendance for exam-related requests will be entertained after the schedule is published.
Posted: 3/22/2026
Mandatory Campus Recruitment Training (CRT) for Final Year
All 4th year (2026 batch) students must attend the Campus Recruitment Training (CRT) programme starting 25th March 2026. Topics covered: Quantitative Aptitude, Logical Reasoning, Verbal Ability, Coding Fundamentals, and Group Discussion / HR skills. Classes will be held every Monday, Wednesday, and Friday from 8:30 AM – 10:00 AM. Attendance is compulsory and will be tracked. Contact the Training & Placement Cell for queries.
Posted: 3/22/2026
Infosys Specialist Programmer Registration Deadline – 20th April
Infosys has announced the Specialist Programmer (SP) drive for 2026 batch students. CTC: ₹9.5 LPA. Eligibility: 60% throughout academics with no active backlogs. Online assessment includes: Advanced Coding (2 problems, 90 minutes) + Computer Science Fundamentals MCQs. Register on Infosys Careers portal before 20th April 2026. The placement cell will share the campus assessment slots once shortlists are received.
Posted: 3/22/2026
Google STEP Internship 2026 – Applications Open for 1st & 2nd
Placement Cell Notice: Dress Code & Conduct During Drives
Scholarship Applications Open – Government Merit-cum-Means
Industry Visit to Tech Park Scheduled – 10th April 2026
Type here to search
30°C Mostly sunny
ENG IN
11:31 AM 3/22/2026

CHAPTER – 5

CONCLUSION AND SCOPE

CONCLUSION:

The AI Resume Analyzer and Job Prediction System is designed to help students reduce the gap between their existing skills and the skills required by the modern IT industry. In today's highly competitive job market, many students find it difficult to understand which technical competencies are necessary for different job roles. This system addresses the problem by providing a centralized intelligent platform that automatically analyzes student resumes, predicts suitable job roles, and guides students toward improving their career readiness.

The platform allows students to create accounts and upload their resumes in digital format. Using Artificial Intelligence and Natural Language Processing techniques, the system extracts relevant information such as educational qualifications, technical skills, certifications, and project experience. Based on the extracted data, the system compares the student's profile with industry job role requirements and performs job prediction analysis.

After analyzing the resume, the system generates a detailed report that highlights predicted job roles along with a confidence score that indicates the level of suitability. It also identifies missing or weak skills required for specific job roles. This helps students clearly understand their strengths and the areas in which they need improvement.

In addition to skill gap identification, the platform provides personalized career recommendations such as suggested technologies to learn, certification programs to complete, and project ideas to practice. This structured guidance helps students focus on relevant learning paths and improve their professional competencies effectively.

Another important feature of the system is the skill authenticity evaluation module, which provides basic assessments to verify the practical knowledge of students in the skills mentioned in their resumes. This helps students build confidence and prepare better for interviews and placement opportunities.

Overall, the AI Resume Analyzer and Job Prediction System provides a comprehensive solution for career planning and employability enhancement. By integrating resume analysis,

job role prediction, skill improvement guidance, and career recommendations into a single platform, the system supports students in making informed career decisions and achieving professional growth in the technology industry.

FUTURE SCOPE

Although the AI Resume Analyzer and Job Prediction System provides several useful features for career guidance and job prediction, there are many opportunities to further improve and expand the system in the future.

One possible enhancement is the implementation of more advanced Artificial Intelligence and Machine Learning algorithms to improve the accuracy of job role prediction and skill gap identification. By analyzing additional factors such as student interests, project performance, assessment results, and career trends, the system can provide more personalized and intelligent career recommendations.

Another improvement could involve integrating the platform with popular online learning services such as Coursera, Udemy, or LinkedIn Learning. This integration would enable the system to automatically suggest relevant certification courses, training programs, and learning resources based on the missing skills identified in the resume analysis process.

The system can also be enhanced by integrating job portal APIs such as LinkedIn, Naukri, or Indeed. This feature would allow students to view real-time job openings that match their predicted job roles and apply directly through the platform. Such integration would make the system more practical and beneficial for placement preparation.

Developing a mobile application for the system is another important future enhancement. A mobile app would allow students to upload resumes, view prediction reports, track skill improvement suggestions, and access career guidance anytime and anywhere, thereby improving system accessibility and user convenience.

Additionally, the platform can be expanded to support multiple colleges, departments, and career domains. This would enable educational institutions to use the system as a centralized career development tool for monitoring student employability and placement readiness.

Future versions of the system may also include advanced features such as automated resume scoring, interview preparation modules, real-time coding assessments, career progress tracking dashboards, and placement analytics reports. These improvements would transform the platform into a comprehensive intelligent career support system.

With these enhancements, the AI Resume Analyzer and Job Prediction System has the potential to evolve into a powerful digital solution that helps students make better career decisions, improve their technical skills, and successfully meet modern industry expectations.

ABSTRACT

In today's competitive job market, students are expected to possess industry-relevant technical skills along with strong academic knowledge. However, many students find it difficult to clearly understand the exact skill requirements for different job roles. This often leads to a gap between the competencies students currently have and the expectations of employers.

The AI Resume Analyzer and Job Prediction System is developed to address this challenge by providing a centralized intelligent platform that analyzes student resumes and compares the extracted skills with industry job requirements. The system performs automated job role prediction and skill gap analysis, which helps students identify missing competencies and understand their current level of career readiness.

Based on the analysis results, the platform generates personalized career recommendations that guide students in learning relevant technologies, completing certifications, and working on practical projects. This structured guidance helps students focus on important skills and improve their professional capabilities in an effective manner.

The system also supports skill authenticity evaluation through basic assessments, enabling students to validate their knowledge and prepare better for placement opportunities. By providing clear insights into strengths and weaknesses, the platform helps students build confidence and plan their career development strategically.

Thus, the AI Resume Analyzer and Job Prediction System acts as an intelligent career guidance solution that assists students in aligning their skills with modern industry demands. By integrating resume analysis, job prediction, skill improvement suggestions, and career planning support into a single platform, the system contributes to enhancing student employability and professional growth.